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Course/Program:- Btech CSE SY

Subject:- Discrete Maths.

Discrete maths Assignment - 2

Q.1) Explain different properties of relation with suitable ex-
Sol:-

1. Inverse Relation: Inverse the order pair elements of relation.

$$\text{ex- } R = \{(1,2), (2,3), (3,4), (4,5)\}$$

$$R^{-1} = \{(2,1), (3,2), (4,3), (5,4)\}$$

2. Empty Relation: A relation are on set A is called empty relation. If set A is empty, so relation of set A is not related to the element set B then it is called empty relation.

$$\text{ex- } A = \{1,2,3\}$$

$$B = \{4,5,6\}$$

$$R = \{(a,b) \mid a+b=99\}$$

$$\therefore R = \emptyset$$

3. Reflexive: A relation R on set A is reflexive if $(a,a) \in R$ for all $a \in A$

$$\text{Ex- } A = \{1,2,3\}$$

$$R = \{(1,1), (2,2), (3,3)\} \text{ is reflexive.}$$

4. Symmetric: R is Symmetric if $(a,b) \in R$ and $(b,a) \in R$.
ex $\rightarrow A = \{1,2\}$
 $R = \{(1,2), (2,1)\}$ is symmetric.

5. AntiSymmetric: $(a,b) \in R$ but $(b,a) \notin R$
ex $\rightarrow A = \{1,2,3\}$
 $R = \{(1,1), (2,2), (3,3), (1,2), (1,3)\}$ is anti-symmetric

6. Transitive: R is Transitive if $(a,b) \in R$ & $(b,c) \in R$
also $(a,c) \in R$
ex $\rightarrow A = \{1,2,3\}$
 $R = \{(1,2), (2,3), (1,3)\}$ is Transitive.

Q.2) Consider the set $A = \{1,2,3,4,5\}$ define a relation
 $R = \{(1,1), (1,2), (2,3), (3,4), (3,5), (4,2), (5,5)\}$ Represent R
using ordered pair

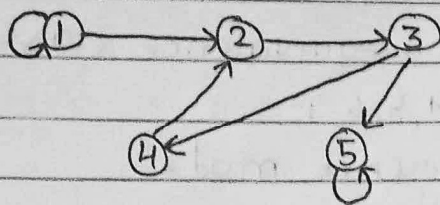
- construct the matrix of R
- Draw the diagram of R .

Solⁿ:

1. Ordered pairs: $R = \{(1,1), (1,2), (2,3), (3,4), (3,5), (4,2), (5,5)\}$
2. matrix of R :

$$R = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}_{5 \times 5}$$

3. Draw the diagram of R:



Q.3) Let $A = \{1, 2, 3\}$ define relation $R = \{(1, 1) (2, 2) (3, 3) (1, 2) (2, 1)\}$
 R = check whether R is reflexive Symmetric Transitive antisymmetric?

Solⁿ:-

1. reflexive: Yes, $(1, 1) (2, 2) (3, 3)$ is present. ✓
2. Symmetric: Yes, $(1, 2) \& (2, 1)$ is present ✓
3. Transitive: Yes, $(1, 2) (2, 1)$ Then $(1, 1)$ is present ✓
4. Antisymmetric: No.

Q.4) Let $A = \{1, 2, 3, 6\}$ Define relation $R = \{(a, b) | a \text{ divide } b\}$
 show that R is a partial ordering relation.

Solⁿ:-

$$R = \{(1, 1) (2, 2) (3, 3) (6, 6) (1, 2) (1, 3) (1, 6) (2, 6) (3, 6)\}$$

1. Reflexive: (a, a) for all present. ✓
2. Antisymmetric: ✓ $(1, 2) \in R$ but $(2, 1) \notin R$ ✓
3. Transitive: $(1, 2) (2, 6) \& (1, 6) \in R$ ✓

Q.5) Define equivalence relation and nequivalence class.

Solⁿ:-

1. Equivalence relation: A relation R on set A is an equivalence relation if it reflexive, symmetric & transitive relation.

2. Equivalence class: If equivalence relation. Then you find equivalence class.

ex $\rightarrow$ if $A = \{1, 2, 3, 4, 5, 6\}$

$R = \text{"Congruence mod 2"}$

Then $[1] = \{1, 3, 5\}$

$[2] = \{2, 4, 6\}$